



Transcranial direct current stimulation influences bilingual language control mechanism: evidence from cross-frequency coupling

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Abstract

How to better suppress the interference from the non-target language when switching from one language to the other in bilingual production? The current study applied transcranial direct current stimulation over the right dorsolateral prefrontal cortex to modulate language control measured by cross-frequency coupling. We found that switching to L2 was more modulated by F4–F3 alpha–beta phase-amplitude compared to switching to L1 after receiving the anodal stimulation at the language task schema phase. These findings suggest that anodal stimulation affects the selection of the target language task schema by enhancing the activation of frontal areas and facilitating the coordination between the left and the right frontal hemispheres.

Keywords Language control · Transcranial direct current stimulation · Language switching · Cross-frequency coupling

Introduction

Bilinguals often switch from one language to the other in response to the communication context. During bilingual language switching, the non-target language needs to be inhibited while the target language is being activated. Thus, the non-target language interferes the activation of the target language and the process of suppressing such cross-language interference is referred as language control (e.g., Baus et al. 2015; Declerck et al. 2015; Gollan et al. 2014; Green 1998). Psycholinguists have been interested in understanding the potential to modulate such language control, namely, bilinguals' ability to switch between languages (e.g., Baus et al. 2015; Costa and Santesteban 2004; Green 1998; Liu et al. 2016), because cross-language communication (e.g., language switching) involves dynamic interactions among a brain network responsible for language control (Abutalebi and Green 2016; Luk et al. 2012). Recently, the transcranial direct current stimulation

(tDCS), a noninvasive and safe cortical stimulation technique, has been used to effectively alter many cognitive functions by changing the polarity of a weak current flow (Fertonani et al. 2010; Nelson et al. 2014; Pope and Miall 2012). Therefore, the current study aims to use the cross-frequency coupling (CFC) to reveal the tDCS effect on language control.

According to the Inhibitory Control (IC) model, during language switching, inhibition works at either or both of the following two phases: (1) the phase of language-task selection (i.e., selecting an appropriate language in reference to contextual cues, e.g., L1, L2); (2) the phase of target-lemma retrieval (i.e., retrieving a target lemma by suppressing the non-target one, e.g., 苹果, apple) (Green 1998). At either or both phases, the amount of inhibition is associated with language activation; that is, the greater the activation, the more the inhibition. The language control mechanism is considered to be utilized during these two phases to inhibit the activation of non-target language task/lemma. Switch costs are considered to be the result of exercising language control at the behavioral level. That is, additional time and effort were occurred when bilinguals switched between their 2 languages (Costa and Santesteban 2004; Costa et al. 2006; Meuter and Allport 1999; Philipp et al. 2007; Philipp and Koch 2009). While behavioral data provide an end point measure of language control, EEG

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data shed insights into the different phases involved during language switching due to their high time resolution. For instance, in the cued picture naming paradigm, we can investigate the control mechanisms locked at the cue phase where bilinguals are instructed which language to name (i.e. selection of the language task), as well as the stimulus phase where bilinguals are instructed to name a picture (i.e. selection of the target lemma) (Liu et al. 2014, 2016, 2018).

When it comes to unbalanced bilinguals whose activation of L1 exceeds that of L2, more cognitive resources are required to inhibit L1 interference during L2 production (Costa and Santesteban 2004; Costa et al. 2006; Green 1998; Liu et al. 2014, 2016, 2018). In the typical language switching picture naming paradigm, for the reason that stronger inhibition on L1 during L2 production carries over to the next trial, it becomes more difficult to overcome such inhibition when switching back to L1. As a result, switching to L1 turns out to be longer than switching to L2 in reaction time measure, showing larger L1 switch costs than L2 switch costs (Costa and Santesteban 2004; Costa et al. 2006; Declerck et al. 2015; Meuter and Allport 1999; Philipp et al. 2007; Philipp and Koch 2009). If bilinguals' ability to switch between languages could be influenced by external drive (e.g., tDCS), we are interested in understanding how this would be applied to reduce/modulate the switch costs during language switching. This would shed light on the extent to which language control is contingent to cognitive control that can be influenced by external mechanisms.

The anodal tDCS (A-tDCS) can depolarize cells and accelerate the transmission of information between neurons. On the contrary, the cathodal tDCS (C-tDCS) hyperpolarizes cells, thus slowing down the transmission of information between neurons (Avenanti et al. 2017; Brunoni et al. 2012; Hill et al. 2016; Meinzer et al. 2012). The sham tDCS (S-tDCS), as control stimuli, gives a shorter stimulation (30 s), but allows participants to have the same subjective experiences as the real stimulation. It affects neither the cell membrane potential, nor the information transmission between neurons. Hence, the A-tDCS may enhance participants' performance, while the C-tDCS weakens. The S-tDCS should produce effects somewhere between the A-tDCS and C-tDCS. However, previous studies demonstrated that cathodal stimulation did not always degrade cognitive performance (Heinen et al. 2016; Pope and Miall 2012; Weiss and Lavidor 2012). Thus, existing researches mainly focus on whether tDCS is an effective stimulation method without dissociating the impact of different types of stimulation. The current study focuses on investigating whether tDCS has an influence on language control and the type of effect will be produced upon stimulation.

Using the classic language switching paradigm to investigate language control (Costa and Santesteban 2004; Costa et al. 2006; Declerck et al. 2015; De Bruin et al. 2014; Liu et al. 2018), we asked unbalanced bilinguals to name pictures in L1 or L2 according to cues. That is, participants were instructed to name a picture either in L1 if cued by a red square or L2 if cued by a blue square, with color being counterbalanced across participants. Previous studies have found that the right dorsolateral prefrontal cortex (rDLPFC) is responsible for suppressing unrelated activation during language switching (Blanco-Elorrieta and Pykkänen 2016, 2017, 2018; De Bruin et al. 2014; Tayeb and Lavidor 2016). Therefore, we applied tDCS over the rDLPFC to examine the effect of tDCS on language control.

In addition, cross-frequency coupling (CFC) was analyzed because CFC reflects the process of transferring information from large-scale brain networks operating at behavioral timescales to the fast, local cortical processing required for effective computation and synaptic modification, thus integrating functional systems across multiple spatiotemporal scales (Canolty and Knight 2010). For example, as Fries et al. (2013) suggested, parietal gamma-band activity was modulated by frontal theta-band activity in successful memory encoding, and this coupling might reflect the reactivation and maintenance of neuronal object representations. Vaz et al. (2017) investigated the potential neural mechanisms of memory coding by CFC. They found that successful memory formation was associated with significant decrease of cross-frequency coupling in left temporal lobe and prefrontal areas, and that such coupling change was observed within various frequency bands. The underlying neural mechanisms: nested oscillations and sharp waveform, were responsible for the changes in these regions. Therefore, the decrease of cortical cross-frequency coupling in episodic memory reflects underlying mechanisms in the human brain. In line with this logic, we employed the cross-frequency coupling analysis to reveal the large-scale network responsible for bilingual language control.

As discussed earlier, unbalanced bilinguals need to suppress stronger interference from the non-target language (L1) when switching to the weaker language (L2). This stronger inhibition could lead to stronger cross-frequency coupling due to information exchange in the frontal network, which is responsible for language control primarily focusing on the couple electrodes located on the frontal and parietal areas (Abutalebi and Green 2007, 2016; Green and Abutalebi 2013; Cooper et al. 2015). CFC is considered an effective analysis approach to reveal the language control mechanism across multiple spatiotemporal scales (Canolty and Knight 2010). Based on the frontal local cortex, we attempt to understand how language control is modified by

effective computation and synaptic modification after receiving the tDCS.

Anodal or cathodal stimulation to the rDLPFC might modulate bilinguals' language control abilities to further affect the inhibition of cross-language interference. The current study investigated this issue by using CFC to understand the brain oscillations among the frontal network responsible for language control during the two phases during language switching: cue-locked selection of the intended language task schema (i.e., language selection) and stimulus-locked selection of the target lemma (i.e. lexical selection). We hypothesize that the A-tDCS and/or C-tDCS affect language control by influencing the coordination between language control brain areas, in particular, that stimulation would exert stronger effects on switching to L2 than switching to L1 because L2 processing is less stable than L1 processing. By contrast, the S-tDCS would not exert influence on the language control neural network, showing no CFC effects. From the theoretical point of view, the current study sheds light on how the frontal network is responsible for working collaboratively in language control to achieve cross-language communication.

Method

Participants

We recruited 31 bilingual undergraduates from Liaoning Normal University, Dalian, China. All of them were right handed with normal or corrected-to-normal vision, reporting neither history of neurological/psychological impairments nor receiving treatment with any psychoactive medication. Prior to the experiment, they signed the written informed consents. The research protocol was approved by the Institutional Review Board at the School of Psychology, Liaoning Normal University. All methods mentioned in this research were performed in accordance with the relevant guidelines and regulations. Data from 9 participants were eliminated due to excessive EEG artifacts. The final sample for analysis consisted of 22 participants (9 male), aged from 18 to 26 years old ($M=21.04\pm2.2$ years).

Table 1 presents the age of L2 acquisition (AOA), the self-evaluated language skills (6-point scale: 1="quite poor", 6="highly proficient") and results from L2 proficiency test. In measuring the self-rated language skills, paired-sample t test indicated that the proficiency ratings of L1 and L2 for all four skills were significantly different: listening, $t(22)=12.09$, $p<.001$; speaking, $t(22)=8.83$, $p<.001$; reading, $t(22)=6.75$, $p<.001$; writing, $t(22)=5.25$, $p<.001$. These results demonstrated that the bilingual participants' proficiency of Chinese and English were

Table 1 Means (SDs) of language proficiency and AOA

	L1(Chinese)	L2(English)
Self-rating		9.00 (2.16)
AOA		
Listening	5.50 (.67)	3.41 (.80)
Speaking	4.86 (.77)	3.18 (.85)
Reading	4.50 (1.37)	2.73 (.88)
Writing	4.36 (1.09)	2.95 (1.36)
LexTALE		36.00 (9.97)
OPT		36.55 (3.97)

significantly unbalanced. Furthermore, they were tested with two objective proficiency tests: LexTALE (maximum score 100 points; Lemhöfer and Broersma 2012) and the Oxford Placement Test (OPT, maximum score 50 points; Allan 1995). The average L2 proficiency was reported in Table 1.

Language switching materials

We adopted the picture-naming task during language switching. The stimuli are 72 line-drawing pictures (15 cm × 15 cm) selected from the picture database of Snodgrass and Vanderwart's photo gallery standardized by Zhang and Yang(2003). The names of all pictures were limited to two-character Chinese words with either one- or two-syllable English equivalents. In order to guarantee similar familiarity between the L1 and L2 picture names, another group of 35 intermediate proficient Chinese (L1)–English (L2) bilinguals who did not participate in the current experiment performed familiarity rating (5-point scale; 1="very unfamiliar", 5="very familiar") on the names of the selected pictures in both L1 and L2. In the average subjective familiarity, no significant difference was found between L1 and L2 (L1: $4.85\pm.08$, L2: $4.84\pm.12$, $t(34)=.69$, $p=.76$).

The selected seventy-two pictures were equally randomized into three sets for each stimulation session (i.e., A-tDCS session, C-tDCS session, and S-tDCS session). Thus, for each participant, there was no repetitive picture across three stimulation sessions. There was no difference in the familiarity of the L1 and L2 picture names in each session (A-tDCS session: $4.86\pm.08$ for L1, $4.83\pm.15$ for L2, $t=.86$; C-tDCS session: $4.85\pm.08$ for L1, $4.86\pm.07$ for L2, $t=.24$; S-tDCS session: $4.84\pm.10$ for L1, $4.86\pm.10$ for L2, $t=-1.04$). A repeated-measure ANOVA was further conducted among the three sessions, and there was no difference between different sessions, either: $F(2, 46)=.36$, $p=.64$.

Procedure

Transcranial direct stimulations (tDCS)

Three sessions, anodal (A-tDCS), cathodal (C-tDCS) and sham (S-tDCS) stimulation, were conducted in our study. The tDCS was administered via saline-soaked surface sponge electrodes (5 cm × 5 cm) by a battery-driven direct current stimulator (ActivaDose II Iontophoresis Delivery Unit, USA). For both A-tDCS and C-tDCS, a constant direct current of 1 mA was applied for 25 min with a ramping period of 10 s at the beginning and the end of stimulation. The active electrode was positioned over rDLPFC corresponding to F4 (international 10–20 EEG system) (Keser et al. 2017) whilst the reference electrode was affixed to the right shoulder with a skin-friendly cello type. For the S-tDCS session, the electrode placement was the same, but the tDCS was turned off after 60 s during the stimulation and turned on for 60 s before the end of stimulation. The stimulation used single-blinded procedure, such that participants were not aware of when and how often each tDCS was applied. At the end of each session, all participants would complete a questionnaire (Fertonani et al. 2010) to assess the Potential tDCS side effect.

The procedure of our study is presented in Fig. 1. The order of stimulation (anodal, cathodal or sham) was fully counterbalanced across participants. That is, equal number of participants were randomly assigned to any of the 6 possible sequences of stimulation sessions: A-tDCS–C-tDCS–S-tDCS, A-tDCS–S-tDCS–C-tDCS, C-tDCS–A-tDCS–S-tDCS, C-tDCS–S-tDCS–A-tDCS, S-tDCS–A-tDCS–C-tDCS, S-tDCS–C-tDCS–A-tDCS. For each session, after stimulation, participants were instructed to do picture naming in either L1 or L2 based on presented cues. In addition, each session was separated by at least 7 days. For example, participant A first received the anodal stimulation and then completed a language switching task in picture naming. 7 days later, Participant A received the cathodal stimulation and completed the second language switching task with a different set of pictures. Again, after

another 7 days, the sham stimulation was delivered to Participant A followed by the final language switching task with the third set of pictures.

Language switching procedure

After receiving tDCS stimulation during each session, participants were encouraged to have the L1 and L2 picture names familiarized in order to minimize naming errors. In the ERP experiment, stimuli were presented at the center of a 17 in. computer screen with a 1024 × 768 pixel resolution. For one trial, a naming cue (red or blue square) appeared for 250 ms initially. The color-language associations were counterbalanced across participants to avoid color preference. Specifically, half of participants used L1 to name the picture if a red square was presented as a cue while a blue square indicated that participants should name the picture in their L2. For the other half, the color-language association was reversed. After the cue, a blank screen appeared for 500 ms, followed by a picture as stimulus. Both naming speed and accuracy were emphasized. The picture would disappear immediately after naming or after 2000 ms upon no response. Finally, a blank screen appeared for 1000 ms before a new trial started.

Two types of trial sequences appeared randomly: switch trials and repeat trials. In the repeat trials, the response language used in the current trial was the same as that of the previous trial (L1L1 or L2L2); whereas in the switch trials, the response language in the current and the previous trial was different (L1L2 or L2L1). There were 5 blocks per session. Each block contained 24 repeat trials of both L1L1 and L2L2, as well as 24 switch trials of L1L2 and L2L1. All stimuli were presented pseudo-randomly. The same picture appeared twice in each block and participants were asked to name the picture in L1 or L2 based on the cue. Participants couldn't predict the following language or picture during the experiment and no more than 3 consecutive trials were named by the same language. Prior to the formal experiment, participants performed 12 practice

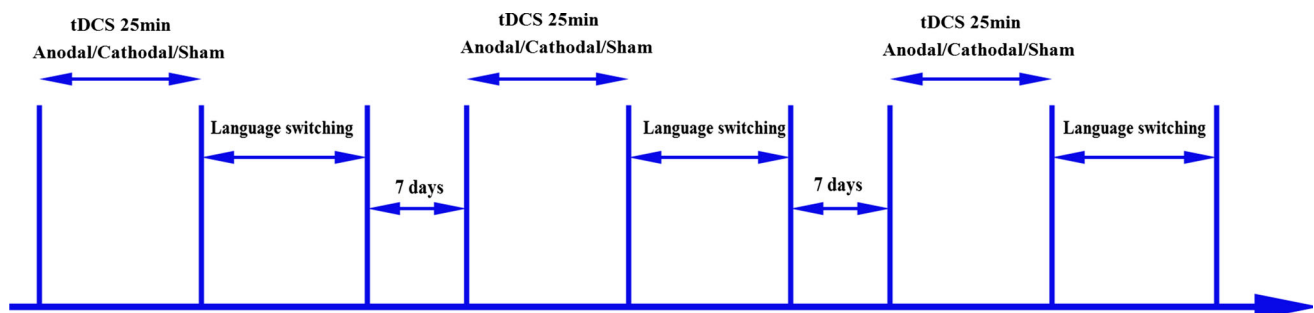


Fig. 1 Experimental procedure. The three sessions of anodal, cathodal and sham stimulation and the language switching task were counterbalanced across participants

trials and their response times were recorded by a PSTSR-BOX connected to a microphone.

Electrophysiological recording

The electroencephalogram was recorded by a set of 64 electrodes from eegmagine (ANT Neuro) according to the extended 10–20 positioning system. During online recording, the sampling rate digitized at 1 kHz and was filtered with a bandpass between .1 and 100 Hz. Electrode impedances were maintained below 5 k Ω across all recording. The signal was referenced to the CPZ electrode. The two electrodes M1 and M2 were placed on the left and right mastoids. For the offline data processing, the EEG activity was referenced to the average of M1 and M2 and re-filtered with low and high cut-off frequencies at .01 and 30 Hz, respectively. EEGLAB (Delorme and Makeig 2004) was used to reduce all types of artifact. ICA component rejection was conducted to reduce ocular artifact and other movement artifacts were visually inspected and manually removed. Epoch of –100–600 ms for cue-locked phase and –100–650 ms for stimulus-locked phase were derived from the continuous electroencephalographic recording. In addition, the pre-cue baseline was 100 ms and the pre-stimulus baseline was 150 ms. The signals in all epochs were selected between –80 and 80 μ V.

Behavioral data analysis

The overall naming accuracy of the naming tasks was quite high (>95%), so it was not included in further analysis. We removed the data from the first two trials in each block, incorrect responses, as well as data from naming latencies beyond $M \pm 3SD$. For naming latency, a three-way repeated-measure ANOVA was conducted with three within-subject factors: tDCS session (A-tDCS, C-tDCS, S-tDCS), language (L1, L2), and language sequence (repeat, switch).

tDCS stimulation questionnaire

We assessed possible stimulation-induced perceived sensations using a translated and adapted version of the standardized questionnaire (Fertonani et al. 2010). The items in these questionnaire were about the sensation influence on participants, including sensory intensity induced by stimulation (5-point scale, 1 = “none”, 5 = “strong”), the span of the influence (beginning = “1”, middle = “2”, end = “3” of the experiment) and the judgments of perceptual sensation (no stimulation/only beginning/only at the end/during the whole experiment/others). In general, the questionnaire was used to evaluate the effectiveness of stimulation. To test the difference between stimulation conditions, one-way

ANOVA was conducted for each sensation item. As shown in Table 2, there was no differences between conditions.

Analysis of cross-frequency coupling

Cross-frequency coupling (CFC) provides a framework for the mechanism of hierarchical nesting between different neural rhythms (see review, Canolty and Knight 2010; Hyafil et al. 2015). There is a correlation between the amplitude of the high-frequency oscillation and the phase of the low-frequency neuroseism (Onslow et al. 2014). Low-frequency neural oscillations may modulate neural connections in long-distance large-scale brain regions with the activation of larger nucleus, while high-frequency neural oscillations are mostly associated with local nucleus activity (Canolty and Knight 2010; Hyafil et al. 2015; Tort et al. 2010). To evaluate the rhythm’s modulation on amplitude, we calculated the modulation index (MI) based on the standard measures previously described in Canolty et al. (2006). The MI was considered a reflection of the coupling between different frequencies which interested us. Based on the standard practice, we divided the frequencies into four frequency bands: delta (1–3 Hz), theta (4–7 Hz), alpha (8–13 Hz) and beta (14–30 Hz) (Burgess 2013; Poikonen et al. 2018; Yun et al. 2012). The classification of high and low frequency is relative: delta and theta frequencies were relatively lower compared to alpha and beta frequencies. We used F1-A (high frequency amplitude) to express the amplitude and F2-P (low frequency phase) to represent frequency phase. The MI calculation procedure was as follows.

1. The eegfilter function in EEGLAB was used to go through band-pass filter, and two different frequency signals were extracted: low-frequency and high-frequency signals (Oppenheim 1999).

Table 2 Means (SDs) of self-rating in the tDCS questionnaire

Items	A-tDCS	C-tDCS	S-tDCS	<i>F</i>
Itchiness	1.77 (.97)	1.64(.79)	1.64 (.79)	1.54
Pain	2.09 (1.02)	2.32 (1.00)	2.09 (1.11)	.68
Burning	1.45 (.67)	1.50 (.80)	1.32 (.57)	.71
Warmth/heart	1.32 (.57)	1.41 (.59)	1.32 (.57)	.32
Pinching	1.41 (.73)	1.45 (.60)	1.32 (.57)	.53
Iron taste	1.00 (.00)	1.05 (.21)	1.05 (.21)	1.00
Fatigue	1.59 (.73)	1.50 (.67)	1.59 (.80)	.24
Effect of performance	1.68 (.57)	1.45 (.60)	1.41 (.67)	1.51
Start	1.55 (.74)	1.82 (.80)	1.45 (.67)	2.41
End	1.68 (.78)	1.73 (.70)	1.45 (.60)	2.56

2. Hilbert Transform was applied to calculate both the instantaneous amplitudes and phase of the signal (Marple 1999; Penny et al. 2008).

$$\zeta t = s(t) + i\bar{s}(t)$$

ζt : raw signal, $s(t)$: real part of the signal, $i\bar{s}(t)$: imaginary part of the signal

$$\bar{s}(t) = H(s) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{s(\tau)}{n - \tau} d\tau$$

$\bar{s}(t)$: the signal transformed by Hilbert

$$A(t) = \sqrt{s(t)^2 + \bar{s}(t)^2}$$

$$\varphi(t) = \arctan \frac{s(t)}{\bar{s}(t)}$$

$A(t)$: instantaneous amplitude of the signal, $\varphi(t)$: instantaneous phase of the signal.

3. The high frequency amplitude and the low frequency phase were combined to create a new complex signal: $Z[t] = A(t)hf. * \exp(i\varphi(t)lf)$.

Z : a new signal, $A(t)hf$: amplitude of high frequency, $\varphi(t)lf$: phase of low frequency

4. The mean of the new complex signal was computed: $z_mean = \text{mean}(z)$, and the mean amplitude was calculated by the MATLAB function 'abs()' to get the 'MVL-MI' (The mean vector length modulation index, $MVL-MI (MI_{raw}) = \text{abs}(z_mean)$).

$$MI_{raw} = \left| \frac{1}{N} \sum_{t=1}^N z[t] \right|$$

5. To attain surrogate data, we had to randomize the instantaneous amplitudes of high-frequency signals (Canolty et al. 2006). The following was how we did this: first, in order to divide the whole time series into two slices, we randomly selected one time point (n) from the time series (old); next, the positions of two slices were exchanged to get a new time series, and a new MVL-MI was also calculated. This procedure ran over and over for one hundred times to calculate the means and standard deviations of all these MVL-MIs.

A normalized MVL-MI was obtained by the raw MVL-MI, the mean and standard deviation of these new MVL-MIs 4136.

$$M = (M_{raw} - \mu) / \sigma$$

M : normalized MVL-MI, M_{raw} : raw MVL-MI value, μ : the mean of these new MVL-MI, σ : the standard deviation of these new MVL-MI

By this procedure, we obtained the cross-frequency coupling. In order to avoid false-positive results, an FDR

(Benjamini and Hochberg 1995; Benjamini and Yekutieli 2001) correction ($q=.05$) was applied for p values of ANOVA in all couples of electronic points in each condition and each coupling frequency band.

Results

Behavioral results

Figure 2 shows the naming latencies and switch costs during bilingual picture naming. The statistical analysis showed a main effect of Language: $F(1, 21)=33.38$, $p<.001$, $\eta^2=.61$, with longer naming latencies for L1 (769 ± 17 ms) compared to L2 (728 ± 14 ms). Also, the main effect of Language sequence reached significance: $F(1, 21)=155.93$, $p<.001$, $\eta^2=.86$, with longer naming latencies in switch trials (760 ± 15 ms) compared to repeat trials (737 ± 15 ms). These main effects were consistent with the literature and our previous studies (e.g., Liu et al. 2014, 2016, 2018).

In order to reveal the symmetry of language switch costs, the subsequent separate analysis by session showed neither significant interaction of Language $\times$ Language sequence for the A-tDCS session: $F(1, 21)=.64$, $p=.43$, $\eta^2=.02$, nor for the C-tDCS session: $F(1, 21)=.07$, $p=.81$, $\eta^2=.03$. This indicates that L1 switch costs (24 ms for the A-tDCS session; 24 ms for the C-tDCS session) were similar to L2 switch costs (18 ms for the A-tDCS session; 26 ms for the C-tDCS session) in both sessions. However, the S-tDCS session showed a significant interaction of Language $\times$ Language sequence: $F(1, 21)=15.4$, $p<.001$, $\eta^2=.41$, such that L1 switch costs (32 ms) were larger than those of L2 (11 ms) (see Fig. 2). In short, we observed symmetrical switch costs for the A-tDCS session and the C-tDCS session, but asymmetrical switch costs for the S-tDCS session. The asymmetry in switch cost after the sham session was typically observed in unbalanced bilinguals, consistent with the literature (Costa and Santesteban 2004; Costa et al. 2006; Declerck et al. 2015; Meuter and Allport 1999; Philipp et al. 2007; Philipp and Koch 2009; Liu et al. 2014, 2016, 2018).

To reveal the differences of switch costs in three sessions, we observed a marginally significant interaction between *Language switch cost* and *tDCS session*: $F(2, 42)=3.12$, $p=.06$, $\eta^2=.13$. A further analysis showed that L2 switch costs in the C-tDCS session (26 ± 20 ms) were larger than those in the S-tDCS session (11 ± 19 ms) only, but did not show any difference between the A-tDCS session (18 ± 20 ms) and the C-tDCS session or between the A-tDCS session and the S-tDCS sessions: $F(2, 42)=3.16$, $p=.05$, $\eta^2=.13$. In addition, L1 switch costs in the A-tDCS (24 ± 26 ms) and C-tDCS (24 ± 28 ms) sessions did not

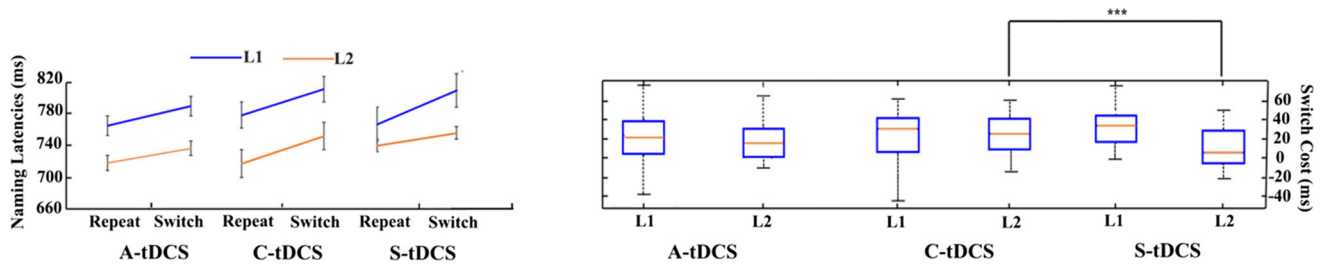


Fig. 2 Mean naming latencies (left) and language switch costs (right) in each session. Error bars for naming latencies show standard errors of naming latencies across subjects calculated separately for each

differ from those in the S-tDCS session (32 ± 18 ms): $F(2, 42) = .86$, $p = .40$, $\eta^2 = .04$. These results indicate that only C-tDCS enabled L2 switch costs to increase, which was neither observed in the S-tDCS session nor the A-tDCS session. At the behavioral level, these findings suggest that cathodal stimulation may hinder the ability to switch from L1 to L2 for unbalanced bilinguals.

Cross-frequency coupling results

(1) Cue-locked different frequencies coupling

Table 3 shows an overview of the main effects of a variable and the interactions between variables. Statistical analyses exhibited a significant main effect of tDCS session, showing that the alpha amplitude was more modulated by the theta rhythm. Pairwise comparisons with Bonferroni correction showed that the modulation occurred in the C-tDCS condition was more significant than that in the A-tDCS condition ($ps < .05$) and the S-tDCS condition ($ps < .05$), and that the modulation in the A-tDCS condition was more significant than in the S-tDCS condition ($ps < .05$). These effects occurred at the 12 dyads of electrodes:

F4–F5, FC2–FC5, FC2–FC3, F1–F6, F2–F5, F2–F1, F6–F5, F6–F2, F6–F6, FC3–FC2, FCz–FC2, FC4–FC2. The main effect of language was also significant at the couple electrodes of FC3–FC1, such that the beta amplitude of L1 trials was more strongly modulated by the delta rhythm than that of L2 trials: $F(1, 21) = 12.36$, $p < .001$, $\eta^2 = .31$.

A significant two-way interaction between *Language* and *Language sequence* was obtained, showing alpha/beta phase-amplitude coupling at the couple electrodes of FC4–F1: $F(1, 21) = 9.73$, $p = .01$, $\eta^2 = .26$. A further analysis showed that L2 switch trials were more modulated compared to L2 repeat trials: $F(1, 21) = 5.86$, $p = .02$, $\eta^2 = .18$.

The three-way interaction of *tDCS session* $\times$ *Language* $\times$ *Language sequence* reached significance at the 2 dyads of electrodes: F4–F3, $F(1, 21) = 8.19$, $p < .001$, $\eta^2 = .25$; F4–F1, $F(1, 21) = 8.71$, $p < .001$, $\eta^2 = .22$. Restricting the analysis just to the variable of *Language*, further analysis showed that the phase of the alpha rhythm modulated the amplitude in the beta rhythm, with stronger modulation occurring in the L2 switch trials than L1 switch trials in the A-tDCS session at F4–F3: $F(1, 21) = 8.19$, $p = .01$, $\eta^2 = .23$; as well as at F4–F1, $F(1, 21) = 8.71$, $p = .01$, $\eta^2 = .24$.

Table 3 Main and interaction effects for cue-locked cross-frequency coupling

Main/interaction effect	Electrodes	<i>F</i>	<i>p</i>	η^2
tDCS session	FC2–FC5	5.19	.01	.16
tDCS session	FC2–FC3	9.64	.00	.26
tDCS session	F1–F6	3.82	.03	.12
tDCS session	F2–F5	8.07	.00	.23
tDCS session	F2–F1	4.67	.01	.15
tDCS session	F6–F5	6.70	.00	.16
tDCS session	F6–F2	4.20	.02	.13
tDCS session	F6–F6	6.08	.00	.18
tDCS session	FC3–FC2	5.53	.01	.17
tDCS session	FCz–FC2	5.28	.01	.16
tDCS session	FC4–FC2	4.76	.01	.15
Language	FC3–FC1	12.36	.00	.31
Language $\times$ Language sequence	FC4–F1	9.73	.01	.26
tDCS session $\times$ Language $\times$ Language sequence	F4–F3	8.19	.00	.24
tDCS session $\times$ Language $\times$ Language sequence	F4–F1	8.71	.01	.23

Similarly, restricting the analysis just to the variable of *Language sequence*, the phase of the alpha rhythm modulated the amplitude in the beta rhythm, with stronger modulation occurring in the L2 switch trials than L2 repeat trials in the A-tDCS session at F4–F3, $F(1, 21)=9.47$, $p<.001$, $\eta^2=.26$ (Fig. 3).

(2) Stimulus-locked different frequencies coupling

Table 4 shows an overview of the main effects of a variable and the interactions between variables. Statistical analyses showed a significant main effect of tDCS session, indicating that the alpha amplitude was more modulated by the beta rhythm. Pairwise comparisons with Bonferroni correction showed that the modulation occurred in the C-tDCS condition was more significant than that in the A-tDCS condition ($ps<.05$) and the S-tDCS condition ($ps<.05$), and that the modulation in the A-tDCS condition was more significant than in the S-tDCS condition ($ps<.05$). These effects occurred at the 5 dyads of electrodes: FC1–FC2, FC1–FC4, FC2–FC2, FC2–FC4, FC4–FC1.

Taken together, at the cue phase, upon the A-tDCS session, we observed an effect of tDCS on language control. That is, switching to L2 was more modulated by the alpha–beta phase-amplitude compared to switching to L1; and such modulation was coordinately achieved by both

Table 4 Main and interaction effects for stimulus-locked cross-frequency coupling

Main effect	Electrodes	<i>F</i>	<i>p</i>	η^2
tDCS session	FC1–FC2	8.90	.02	.14
tDCS session	FC1–FC4	5.30	.01	.16
tDCS session	FC2–FC2	6.67	.00	.20
tDCS session	FC2–FC4	4.24	.02	.14
tDCS session	FC4–FC1	4.20	.02	.13

right and left frontal areas (e.g., F4–F3, F4–F1). By contrast, the behavioral results showed increased L2 switch costs upon the C-tDCS session compared to the S-tDCS session; however, this modulation in CFC was absent. More importantly, both anodal and cathodal stimulation showed symmetrical language switch costs (i.e. switching to L1 and L2 incurred similar costs). What is typically observed in language switching is that L1 switch trials incurred more cost than L2 switch trials for unbalanced bilinguals (e.g., Costa and Santesteban 2004; Costa et al. 2006; Declerck et al. 2015; Meuter and Allport 1999; Philipp et al. 2007; Philipp and Koch 2009; Liu et al.

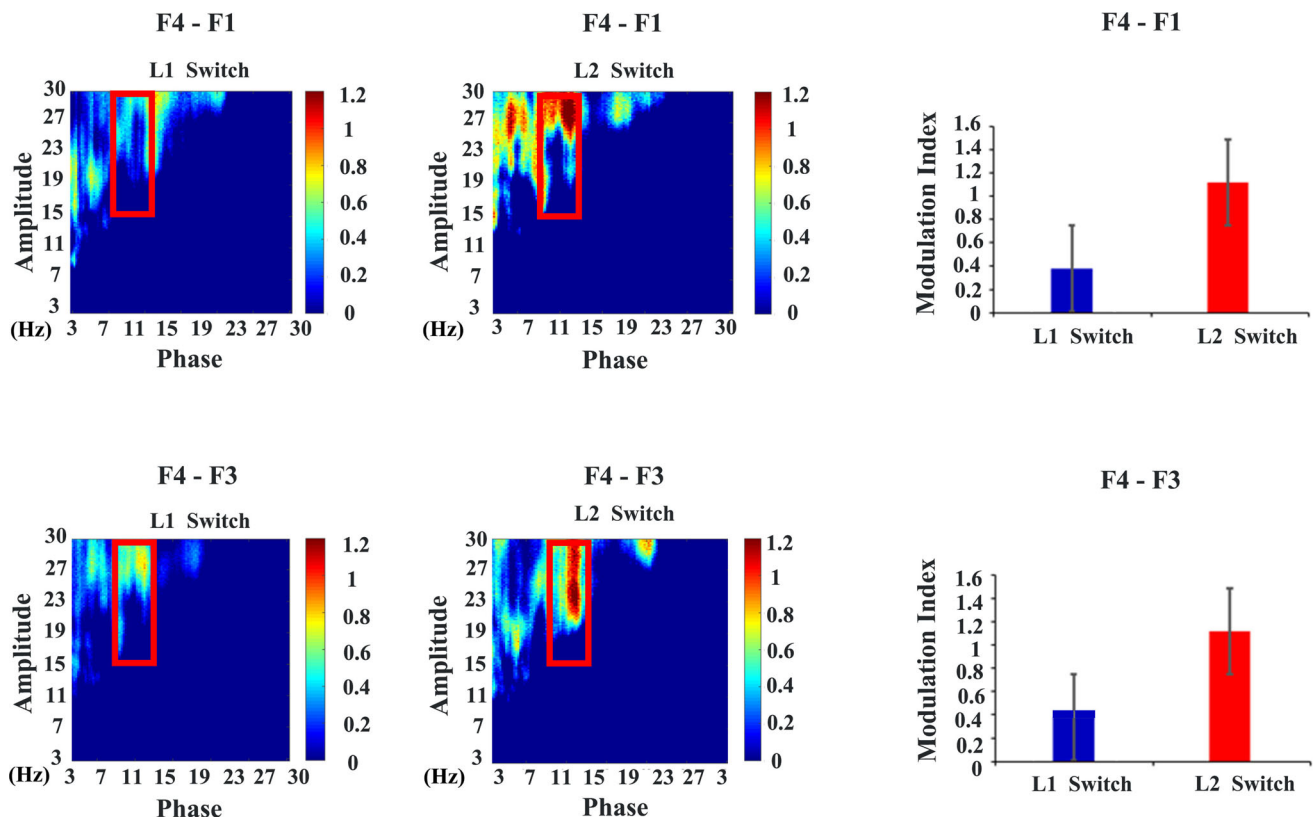


Fig. 3 Cross-frequency coupling at the cue-locked phase. In the A-tDCS session, the alpha rhythms of L2 switches show stronger modulation on the beta amplitude of L1 trials at 2 dyads of electrodes:

F4–F1, F4–F3. The red-line boxes mark the focused frequency bands. (Color figure online)

2014, 2016, 2018), like what we observed in the S-tDCS session. In a nutshell, tDCS produced an effect on language control so that asymmetrical switching costs in the S-tDCS session became symmetrical in both A-tDCS and C-tDCS sessions. That is, switching to L2 induced more inhibition than switching to L1 upon stimulation. This effect was consistently observed in L2 switch trials, instead of L1 switch trials, in both types of stimulation; while A-tDCS showed this effect only in the CFC results and C-tDCS showed this effect only in the behavioral results.

Discussion

The current study investigated the effect of tDCS on language control by applying tDCS over the rDLPFC during language switching. To summarize our main findings, at the language task schema phase (i.e., cue processing), the A-tDCS exerted an effect on language control, that is, switching to L2 was more modulated by alpha–beta phase-amplitude compared to switching to L1, and such modulation was coordinately achieved by both right and left frontal areas. By contrast, the C-tDCS failed to show any significant CFC effect while switching to L2, instead, increased L2 switch costs were observed at the behavioral level in C-tDCS. These findings suggest that tDCS influenced language control only when unbalanced bilinguals switched to their weaker language (i.e., L2).

The advantage of the EEG method is to provide highly time-sensitive measure of brain activity corresponding to cognitive functions observed in a task. The significant CFC observed in L2 switch trials in the A-tDCS condition indicated an early effect of inhibition when selecting the target language task schema (i.e., the cue-locked phase). Stronger CFC indicated increased brain oscillations among the frontal network (F4–F3) that is responsible for language control and subsequently led to more efficient inhibition when unbalanced bilinguals switched to L2, compared to switched to L1 or stayed non-switched in L2. This inhibition was captured during the early phase of selecting the language task, but not reflected in the behavioral data. As CFC serves as an indicator of coordination between brain areas in cognitive processes, we can also speculate that increased ability of language switching upon the A-tDCS at the early language task selection phase changed the pattern of switch costs from asymmetry to symmetry. In other words, language control of unbalanced bilinguals was influenced so that they behaved similar to balanced bilinguals.

Our main finding is the increased brain oscillations at F4–F3 when bilinguals switched to L2 to enhance language

control ability upon the A-tDCS. This result was demonstrated in CFC, a novel approach to understand brain oscillation in a cognitive task. To be more specific, switching to L2 showed more modulation by the alpha–beta phase-amplitude. In the literature, alpha activity is considered to be associated with the inhibition of task-irrelevant networks (Klimesch 2012; Benedek et al. 2014; Payne and Sekuler 2014). It thus plays an important role in limiting attention to the target stimuli and blocking processing of the irrelevant events (Klimesch 2012). By this logic, our bilingual participants needed to inhibit the interference from the non-target language task schema of the previous trial, thus, more inhibition was required when switching to L2 than switching to L1, as inhibiting L1 was more effortful when bilinguals switched to L2. In addition, beta activity represents the active maintenance of the current network configuration (Lewis and Bastiaansen 2015; Klepp et al. 2015; Pastötter et al. 2008). Bilingual participants also needed to maintain the target language task schema to name pictures in the subsequent lemma phase. Taken together, the inhibition-related alpha activity in combination with the maintenance-related beta activity coordinately completed the selection of the target language task schema, showing stronger cross-frequency coupling in the alpha–beta amplitude-phase while switching to L2 than switching to L1.

Our finding also suggests that the effect of increased brain oscillations at F4–F3 was not just restricted to the brain area that was directly receiving A-tDCS. We only stimulated the rDLPFC, but found cross-frequency coupling at both left and right frontal electrodes (F4–F3). Previous evidence suggested that the bilateral inferior frontal gyri (i.e., the left and right inferior frontal gyrus) are responsible for suppressing automatic processes and inhibiting interference from the irrelevant information (such as prepotent responses from the dominant language) (Abutalebi and Green 2016). Specifically, the left prefrontal cortex is more related to response selection and its right counterpart is rather associated to response inhibition (Aron et al. 2007). In the current study, although tDCS was applied on the right frontal electrodes, our CFC effect suggested that brain oscillations also travelled to the left frontal area, thus leading to the increased brain oscillations of the language control neural network. In short, A-tDCS affected the process of selecting the target language schema by enhancing the activation of the frontal areas to impact the coordination between the left and right frontal hemispheres.

While we failed to observe any CFC effect of language control in the C-tDCS condition, a significantly increased L2 switch cost emerged, compared to the S-tDCS

condition. The C-tDCS was reported to inhibit the activity in the cortical area in the literature (Fregni et al. 2005; Avenanti et al. 2017; Brunoni et al. 2012). As a result, applying C-tDCS induced more difficulty when switching to L2, compared to that in the S-tDCS condition. This increase in L2 switch cost also changed the pattern of switch costs for our unbalanced bilinguals, as they behaved similar to balanced bilinguals showing symmetry in switch costs upon C-tDCS. Thus, the symmetry observed in the C-tDCS session was driven by a different brain mechanism from that observed in the A-tDCS session, although both types of stimulation altered language control in unbalanced bilinguals.

As anodal and cathodal stimulation enhanced or reduced the frontal activity, why did L1 switch trials fail to generate pronounced behavioral/neural differences across the C-tDCS, the A-tDCS and the S-tDCS sessions, unlike L2 switch trials? We reason that unbalanced bilinguals are more automatic in processing their native languages (L1) and L1 processing would not be affected by temporary excitation or inhibition on the frontal areas. Therefore, there was little difference in L1 switch costs in both A-tDCS and C-tDCS sessions compared to the S-tDCS sessions. Another possibility is that L1, located in the left inferior frontal gyrus (Broca's area 44, a critical region for speech production), would not be affected by the stimulation on the right frontal areas (Cantalupo and Hopkins 2001; Watkins and Paus 2004). Thus, this stimulation didn't affect L1 processing. Future study could devote to resolving the above issues by investigating the tDCS effect on balanced bilinguals to minimize L2 proficiency confound.

Conclusion

The current study applied tDCS to influence language control by using cross-frequency coupling analysis. We found that both left and right frontal areas worked collaboratively to select the target language task schema during bilingual language switching under the anodal stimulation, when unbalanced bilinguals switched from L1 to L2, but not from L2 to L1.

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